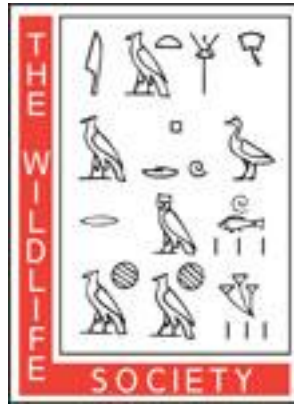


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Deer and Dauerwald in Germany: Any progress?

Hans G. Schabel

Abstract In 1936, Aldo Leopold published 2 landmark papers on history, ecology, and policy issues concerning "Deer and Dauerwald in Germany." His observations were based on several months of studying game-forest conflicts in Central Europe. On the one hand, he returned highly critical of a land-use system in which silviculture and deer management were on constant collision course. On the other hand, he lauded the newly mandated Dauerwald concept, which promised peace in the forest based on a holistic land ethic. This paper briefly characterizes the Dauerwald, then examines the demise, dormancy, and revival of the Dauerwald since Leopold's visit 65 years ago.

Key words Dauerwald, deer management, forest history, nature conservation, restoration, silviculture

In 1935, Aldo Leopold studied game and forest management in Germany. His visit spawned 6 papers (Leopold 1936*a-e*, 1991). Two of these (1936*a,b*) contrasted the long history of mutual interference between game and forestry in Germany with the newly mandated but ultimately short-lived "Dauerwald" (DW) or "permanent forest," which promised greater harmony through a radical reform of game management and silviculture.

Half a century later, Wolfe and von Berg (1988) again investigated game-forest conflicts in Germany. They concluded that the basic elements of the deer-forestry problem described by Leopold 50 years ago remained largely unchanged, in fact were perhaps even accentuated, because of increased deer densities and continued spruce (*Picea abies*) monocultures.

Since then, Dauerwald-like policies have been reinstated in much of Central Europe (Schabel and Palmer 1999). At the same time, deer densities in Wisconsin and much of the United States (U.S.) are the greatest in 100 years (Meadows 2000) and have reached or exceed social carrying capacity. This is creating a crisis (Waller and Alverson 1997) reminiscent not only of previous German conditions but also of one that had already greatly frustrated Leopold during his tenure with the Wisconsin Con-

servation Commission from 1943 to 1948 (Flader 1978, McCullough 1987). As a result, this may be the proper time to revisit relevant issues and examine what progress may have been made in Germany during the last decade in reconciling deer with silviculture. This will be done by first tracing the history of game and forests in Germany, then by characterizing the DW, and finally by focusing on the present status and trends.

Pre-Dauerwald

In his first DW paper, Leopold (1936*a*:366) took the reader step by step through "the rich alluvium of German (forest) history," outlining distinct periods which ultimately led to the adoption of the DW idea. According to this account, hardwoods and hardwood-conifer mixes prevailed during the period of the "Urwald," or primeval forest, when game populations were presumably stable and high. During the "Feudal Forest" period of 1100 to 1400, hunting changed from industry to sport. Some of the more vulnerable megafauna, like aurochs (*Bos primigenus*) and wild horses (*Equus gmelini*), were hunted to extinction during this time, exotic fallow deer (*Dama dama*) were introduced, and mast-producing oak (*Quercus* spp.) and beech

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(*Fagus sylvatica*) were favored for the benefit of deer, wild boar (*Sus scrofa*), and domestic swine. From 1400 to 1618, the “Big High” resulted in such numbers of deer as Germany had never seen and will never see again. This deer glut was for the exclusive benefit of clergy and nobility, with game management provided by so-called “foresters.” Forest composition during this period was still more or less natural. As a result, despite large numbers of game, the first records concerning game damage appear only toward the end of this period. This spoke volumes to Leopold about the carrying capacity of the hardwood forest.

The Thirty Years War precipitated an era of “Exploitation” lasting until about 1700. Deer suffered serious declines, brown bears (*Ursus arctos*) were exterminated, and forests were overcut, overgrazed, and overraked. The ensuing “Timber Famine” during the eighteenth century was a period of many deer and few forests, leading ultimately to cellulose forestry to correct the timber shortages. “Spruce mania,” and in places a Scots pine (*Pinus sylvestris*) equivalent, coincided with a sharp decline in deer numbers until about the turn of the century. This reduction in deer populations was in part due to poor habitat, but was also attributable to the fact that hunting restrictions were dramatically liberalized after the 1848 Revolution. Management of spruce and pine generally relied on planting and clear-cutting of monocultures, a system that aligned perfectly with the militarily mechanistic and systematic *Zeitgeist* and the concurrent spirit of classic liberal economic theory that emphasized maximization of profit. This policy of rational forestry did in fact achieve the objective of alleviating timber shortages. For lack of suitable habitat, however, the rebuilding of deer numbers toward the end of the nineteenth century was possible only by winterfeeding, predator suppression, various mechanical measures for tree protection, and compensation of landowners for game damage suffered. This simultaneous rise of deer and conifers was, in Leopold’s opinion (1936b, 1991), accomplished at the expense of the health of an already strained ecosystem and therefore of the public interest.

Dauerwald

Disenchantment with the newly created “wood factories” (Leopold 1936b:464) was expressed as early as 1824 by von der Borch and later by König

(1849), who deplored the effects on landscape beauty and forest health. Shortly afterward, a return to more natural forests was demanded, most prominently by Gayer (1886), at a time when the owner of a forest estate at Bärenthoren in the state of Saxony-Anhalt had already begun implementing such change. He, like other landowners, had started to pay the price for age-old land abuse, including repetitive monoculturing of conifers, as reflected by soil fatigue and declining yield, landscape and biotic monotony, and constant battles with pests and other stresses.

The Bärenthoren model caught the attention of a scientist, Alfred Möller. His professional background in forest mycology, influences by Gayer and others, as well as impressions gained in Bärenthoren and in virgin forests of North and South America, helped him synthesize a conceptual framework for his brainchild, the DW. In his book, Möller (1922) outlined this new philosophy, which envisioned permanently stocked, productive, and qualitatively superior forests that satisfied the full range of utilitarian, environmental, and social expectations. In his opinion, the DW not only respected the integrity of soil and indigenous biodiversity but also acknowledged intergenerational responsibility. He viewed forests as complex organisms that can express their inherent vigor and productivity only “if all parts are healthy” (Möller 1922: 39), obviously arguing a land ethic that may have inspired Leopold’s (1953:147) “first precaution of intelligent tinkering”—i.e., the keeping of “every cog and wheel.”

Möller’s rightful assertions that implementation of the DW would be a lengthy and possibly a painful process did not help his cause. Especially his call for a reduction in deer densities and an abstention from clear-cutting pitted him against entrenched interests and complacency and produced often vitriolic discussions between recalcitrants and reformers (later classified by Leopold as “type A and type B” foresters, respectively) for years. Ultimately, however, the latter prevailed. In 1934 the DW was mandated for all forests in Germany.

During the following year Leopold and 5 other American foresters happened to visit this country. While Leopold was impressed with the exemplary system of farm game management (Leopold 1936c), the newly begun “very positive and aggressive program of wildlife restoration” (Leopold 1936e:102), the Germans’ institutional order

(Leopold 1936*d*), as well as their "vigorous esthetic discontent" (Leopold 1991:227) and "nostalgia...for wildness" (1936*e*:102), he rated the condition of migratory and forest game as "in a bad way" (Leopold 1936*c*:86 and 1936*d*:67, respectively). He scathingly criticized the "over-artificialized" (Leopold 1936*e*:108) "engineering era" (Leopold 1936*e*:110), "cabbage brand" (Flader and Callicott 1991:271) or "wood-factory" (Leopold 1936*b*:464) silviculture, and a "cubistic forestry" (Leopold 1991:227) afflicted with an "epidemic of geometry" (Leopold 1936*e*:111), which together with excessive deer herds had been, in his opinion, ruinous to many species of songbirds and flowers among other indicators of landscape quality. At the same time, he expressed hope that the DW mandate would eventually undo this "burglary in esthetic wealth" (Leopold 1991:228), as already evident in attitudes and practices of some landowners he had met (Leopold 1936*e*, 1991; Trommer 1996).

All together, Leopold (1936*a*:366) felt his analysis of old versus new forestry to "be rich in lessons for all countries whose conservation policy is still in the making," including the U.S., and he made appropriate recommendations (1936*b*). The German experience, like other events in 1935, made this a landmark year in Leopold's life. His eye-opening visit heralded a reorientation in his evolving systemic worldview or a "personal transmutation of values" (Flader and Callicott 1991:239) and challenged some of his most basic assumptions about the ultimate possibility of environmental control.

Dauerwald since Leopold

Demise and dormancy

The winds of change—i.e., the retreat of the conifers and a return to nonviolent forestry and to a gentler and more restrained land use which Leopold (1936*e*; 1991) optimistically felt had begun with the DW mandate—unfortunately turned out to be just a gust. Only 3 years after its inception, when Germany started preparing for war, political and hunting interests conspired to abolish the DW policy. In the early 1950s, however, a "Working Group for Nature-friendly Forest Management" (ANW) revived the DW movement under a different name, probably because of the ideological stigma associated with the DW's often bitter debate and its original adoption by the Third Reich. Interestingly, support for the ANW originated largely with large private forest owners attracted by the

DW's promise for long-term cost-efficiency.

In 1951, DW policies were officially revived in the German Democratic Republic (DDR) (Thomasius 1990). As previously in the Third Reich, however, this second attempt to force a silvicultural revolution was short-lived. Among other reasons, excessive deer herds in this "hunting paradise," which had been rebuilt after WW II, were simply incompatible with this effort. The emerging deer mania served in part to satisfy perks for communist party insiders, in part to bolster a financially strained system by generating Western hard currency through hunting tourism. Gold-medal deer trophies brought recognition by supposedly reflecting a superior system, in the same way that Olympic gold amassed by the DDR's carefully nurtured sports elite did. Ten years later, highly mechanized, large-scale clear-cuts and industrialized production forestry became the national standard in the DDR, complementing the collective megafarms whose blandness already marred the landscape. Thus, in a territory that had once been the domain of hardwoods (97% of the land), 77% of forests eventually were stocked in conifers, 54% of them Scots pine monocultures. Altogether, 61% of forests there were composed of sticks less than 60 years old (Thomasius 1990, Anonymous 1994*a*).

In West Germany, DW ideas remained largely dormant until the late 1980s, except with some ANW enthusiasts and in parts of southern Germany, where a quiet "unofficial" revolution to "seminatural forestry" in state forests had begun in the 1970s (Spathelf 1997). Overall, though, as in the east, young forests largely composed of conifers continued to dominate, two-thirds of them having originated since WW II (Sperber, in press). Driven in part by huge clear-cuts made by occupant forces (Anonymous 1994*a*), the silvicultural system still relied on even-aged monocultures of conifers in 40% of forests and concurrent widespread use of clear-cuts followed by artificial regeneration (Wolfe and von Berg 1988, Plochmann 1992). Spruce and pine now covered 66% of the forest landscape, a complete reversal of the presumably natural 2:1 hardwood-conifer ratio. Individual districts had even experienced increases from originally only 20% to more than 90% conifers (Plochmann 1992).

Forestry in crisis

Germany's persistent conifer glut coincided with an unprecedented monoculturing of deer once the herds had recovered from a postwar low. Their

management increasingly relied on practices suggestive of game ranching (Geist 1985, Sperber 1994). Hunters and hikers alike enjoyed frequent encounters with game. Attempts to effect change were, however, evident. In the late 1960s, for instance, maverick foresters in Bavaria triggered a brief “roe deer war,” and in 1971, a TV show exposing red deer (*Cervus elaphus*) management practices stirred the ignorant public in unprecedented fashion (Sperber 2000). A resolution passed by 112 concerned German forest scientists in 1974 called for abolition of the contemporary hunting system, but the edict remained unheeded due to the solidly entrenched hunting interests of leading segments of society (Sperber 2000). Thus in the late 1970s and early 1980s, as many as 100,000 red deer and more than 2 million roe deer (*Capreolus capreolus*) were estimated to exist in West Germany, a country roughly the size of Oregon. At ungulate game densities estimated at 6.7 animals/km², Germany ranked second in Europe in this category (Ueckermann 1987, Wolfe and von Berg 1988).

According to Wolfe and von Berg (1988), these figures translated to an average density of 3 red and 8 roe deer/100 hectares, respectively. However, summer densities of 20 to 30 roe deer/100 hectares of forest and winter densities possibly twice that level were common in certain areas. These figures contrasted with economic carrying capacity estimates ranging from only 1 to 1½ red deer /100 hectares or 4 to 5 roe deer/100 hectares for sites of low to moderate habitat quality, as would likely occur in closed monocultures with few openings and little interspersed of agricultural lands (Ueckermann 1987).

All the while, a new storm was brewing in the form of “Waldsterben,” a complex forest decline that today afflicts almost two-thirds of trees in Germany (Anonymous 1997, Kändler 2000). This disorder is primarily attributable to air pollutants (Waldsterben from above) in conjunction with a presumably pollution-driven heating-drying of the climate. Various other stresses and conditions, however, such as soil malaise, pest and weed problems, and continuing species losses, were considered contributors to deteriorating forest health. Their effects were directly or indirectly linked, at least in part, to traditional forest and game management practices or to more recent influences, such as excessive reliance on pesticides and heavy machinery and increasing recreational pressures (Anonymous 1994a). Deer specifically were accused of being players in Wald-

sterben from below, by impoverishing the ground flora (Sperber 2000), a connection already implied by Leopold (1936b,e). According to Ziegler-Hennings (1991), Germany's 1984 red list included as endangered 32% of ferns and flowering plants, and 80 to 90 indigenous plants and animal species were estimated to disappear in this country annually. Furthermore, 38% and 72.9% of the sapling population of conifers and deciduous trees, respectively, exhibited game damage in West Germany in the late 1980s (Anonymous 1990). Red deer even severely debarked poles of spruce, a species generally ranking low in palatability (Ueckermann 1987). Total cost for deer damage and for prevention measures approached (US) \$200 million for West Germany alone, accounting for 80% of forest protection budgets (Ueckermann 1987) and an estimated 10% devaluation of the total net production of forestry (Speidel 1980).

This forest health crisis conspired to catalyze 2 separate, concurrent trends, a persistent economic crisis in forestry and an increasing “greening” of a nature-hungry German public and of many foresters. As a result, Wolfe and von Berg's (1988:31) predictions “that public pressure will ultimately result in legislation that will bring about modifications in deer densities” and that “this may permit implementation of a more natural system of forestry” were correct and timely. In 1987, the state of Saarland terminated clear-cutting and several other German states and countries in Europe followed suit by mandating or promoting natural forest management—essentially DW systems—for public forests. The “century storms” of 1990 (Anonymous 1994b) and of 1999, which most severely affected spruce plantations, idiosyncratically and yet very timely accelerated this transition, and so did legal commitments made at the Rio Earth Summit.

Reform

If Leopold were to revisit Germany today, what would he observe and how would he likely judge the situation? He would probably first express his amazement at how long it took to reinvent the DW. He would then undoubtedly be pleased to learn that forest administrations in all German states have now officially committed to and are in the early stages of implementing natural forest management systems, though to various degrees of purity in different states as judged by nature conservation groups (Späth and Plieninger 1996). The fact that



Figure 1. The current de-emphasis of conifers and transition to more hardwoods is symbolized by this birch seedling germinating in a pine stump. Depressed deer populations are one major strategy to effect such change. Early in his career, Leopold categorically favored pine over birch, but the Germany trip taught him much greater appreciation for hardwoods. Photo by H. Schabel.

Leopold himself had experienced difficulties in bringing about change in resource policy and practice, and even more so in effecting attitudinal shifts in diverse stakeholders, would allow him to realistically assess Germany's likely speed and progress in moving from sustainable yield of wood and deer to sustainable forestry (Anonymous 1997) in the broadest sense.

The present transition to DW (Figure 1), by whatever name, is on the one hand marked by de-emphasis of deer, pine,

spruce, and exotics and on the other by remedial actions to assist uncommon or rare indigenous biota in recapturing their proper niches, irrespective of their direct commercial value (Schabel and Palmer 1999). Such objectives acknowledge the "deep respect for natural mixtures and deep suspicion of large pure blocks of any species," that Leopold (1936b:465) had recommended as the third of his 5 "specific amendments of American practice...indicated by the German experience." They also recognize the ecosystem value of biodiversity, as demanded by Rio agreements. The long-term result envisioned is the restoration of near-natural but managed forest communities that approach their inherent, site-specific potential for complex structure, composition, and resilience. Implementing such an ambitious program presumes alignments in the often difficult relationship between foresters, game interests, and nature conservation groups (Sperber, in press). Although some rapport among these constituents may have developed, the discourse often seems more animated than amiable.

Game and hunting. As anticipated, the DW transition will be a generational effort that may entail sacrifice. Among the first to pay for implementation of the new policies were some hunting interests. To facilitate natural or artificial restoration of a diverse, native flora and associated fauna, deer densities were cut back significantly, at least on state lands, as reflected in recent harvest data (Figure 2). This indicates that at least for the time being, ungulate game in at least one-third of German forests is subordinated to the goal of vegetational recovery ("Wald vor Wild" or "forest before game")—that is, a shift from game to habitat.

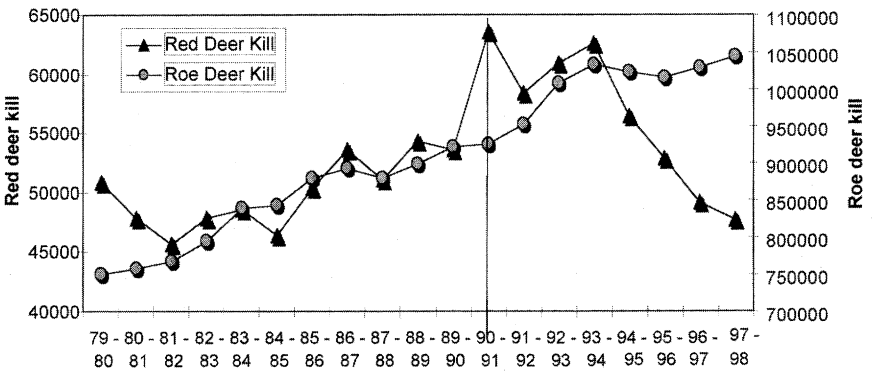


Figure 2. Annual harvest (1979 to 1998) of red deer (*Cervus elaphus*) and roe deer (*Capreolus capreolus*) in the Federal Republic of Germany (after German Hunting Club 1999). Reunification in 1989 coincides with a particularly sharp decline in red deer, populations of which had been excessive in East Germany. Note different scales for the two species.

Incidental to this development came several other changes in established practice. For instance, the traditional, selective shooting of roe bucks for trophy management purposes was abandoned in those districts where deer reduction was being pursued. Even age-old, almost sacred hunting privileges of the forestry profession, a highly valued and jealously guarded fringe benefit, were curtailed in several states. In Bavaria, for instance, foresters are now required to purchase licenses to pursue trophy animals (Sperber, in press), and in Baden-Württemberg, a number of state hunting districts were recently leased out for the first time. Leopold (1936b:463), never one to gloss over inconvenient truths, had implicated some German forestry colleagues in conspiring in the maintenance of abnormally high deer densities and in being “unwilling to curtail their private pleasure for the public good,” at the same time that he accused the Germans’ general “love affair” with deer. To gauge the effect of deer reduction, the traditional mode of assessing densities by game censuses, a procedure that was very subjective and conveniently allowed underestimation of numbers, has now been supplemented by additional indicators. Signs of health in individual deer and periodic browse surveys of terminal shoots in shrubs and young trees are considered more reliable (if still imperfect) measures of deer density and impact (Kech 1997). Based on such surveys, more than two-thirds of public forests in Baden-Württemberg are now considered to have accomplished satisfactory deer reduction (H. Weiss, Baden-Württemberg Forest Service, personal communication), and in neighboring Bavaria, “distinctly more favorable forest regeneration on state forests” has been achieved (Sperber, Bavarian Forest Service, personal communication). At the same time, the reductions and purported insularization of red deer in some areas have prompted concerns for the development of genetic bottlenecks in certain red deer herds (H. Kalchreuter, European Wildlife Research Institute, personal communication).

Even so, game densities are still considered locally too high (Sperber, in press). This is particularly true on private or community lands often leased to nonlocal hunters. But even in state forests of Rhineland Palatinate, for instance, game-damage-prevention costs still amount to 24 DM (over US \$13)/hectare/year (Ecological Hunting Club 1999). In the same state, 60% of roe deer and 45% of red deer hunting districts do not yet satisfy silvicultural criteria for compliance with legal requirements, as

interests of landowners and hunting-rights owners are still in conflict (Schaefer 2000). In most states of the former DDR, deer densities to this day exceed those of the West (Sperber, in press), remaining far from Leopold’s (1936b:465) ideal of a “generous policy in building carrying capacity and a stingy one in building up stock” or a “comfortable margin of safety between the two.”

Following DW mandates and some initial adjustments in hunting practices and state laws, efforts are now underway to effect more sweeping changes in the traditional hunting system. In 1988, an Ecological Hunting Club (ÖJV) was founded, challenging the long-established monopoly of the numerically and otherwise powerful German Hunting Club (DJV), which includes 290,000 members or 86% of German hunters (A. Keulen, German Hunting Club, personal communication). Most of the 3,000 members of the ÖJV are reformed foresters and hunting landowners with a proclaimed “Wald vor Wild” philosophy. As James I of England, who valued trees over game, had already learned a long time ago, such emphasis is certain to be unpopular (Fisher 1895). As a result, it may not surprise that some traditional hunters contemptuously dismiss the young ÖJV as a “small bunch of vocal extremists.” Leopold himself instantly earned a substantial dose of resentment when he fought hunter opposition to reductions in excessive deer herds in Wisconsin and “was personally cast as the scapegoat for the crime of 1943,” when an attempt was made to cut back deer populations for ecological reasons (Flader 1978:202).

Presently an overhaul of German federal hunting laws is in discussion, which may result in more changes in hunting policies and practices (G. Sperber, Bavarian Forest Service, personal communication). While some argue that no change in these laws is needed, given recent progress in silvicultural reform (K. Kalchreuter, European Wildlife Research Institute, personal communication), the increasingly powerful German Alliance for Nature Protection (DNR), which reflects a broad coalition of diverse interests, is pushing change with far-ranging proposals including suggestions for redefining hunting objectives, reassessing protected-species listings, shortening hunting seasons, increasing landowner participation in hunting, prohibiting trapping, restraining the release of non-indigenous game, reforming hunter education, outlawing winterfeeding, modifying hunting methods, mandating browse surveys for decision-making, and reducing

the trophy emphasis, among others (Bode and Emmert 1998, Anonymous 2000). Heated discussions are almost assured, as issues will touch not only on resource management but also on questions of philosophy, ethics, and social politics, for the German hunting system is less egalitarian than the North American one. Sperber (1994), himself a forester and hunter, echoes the increasingly vocal nature of the conservation movement in his accusation that the German hunting fraternity is a "neo-feudal cult." Irrespective of which agreements may materialize from these discussions, the core issue for reform of the prevailing hunting system remains to eventually assure satisfactory natural forest regeneration "without auxiliary means" (Schaefer 2000:296). A secondary result may well be a "new" or at least reformed hunter, as demanded by Bode and Emmert (1998), an individual less driven by prestige-based trophy hunting than by a passion for hunting as a nature appreciation and an applied conservation tool in the context of a broader commitment to the restoration of native flora and fauna. While the ÖJV already sees itself in that ambitious image, large segments of the DJV seem averse to or slow in embracing the transition demanded by reformers (G. Beyer, German Nature Conservation Coalition, personal communication).

Thus, after first being an industry and then a sport, hunting is now being cast by reformers as an ecological tool to assist nature in a severely altered, man-made environment. This by necessity rearranges the priorities from managing game largely by empirical means to managing wildlife based on science. The 1987 Federal Nature Protection Act actually facilitated this shift (Wolfe 1991). Leopold, as the U.S.'s first professor of wildlife management, would consider this development long overdue. He also would be humbled to learn that the newly created professional Society for Wildlife Ecology and Nature Conservation (Figure 3) was named in his honor "Aldo Leopold Gesellschaft" (www.ALGnet.de) and shares his unabashed criticism of the German game-forest dilemma. Leopold had lauded a similar effort at nature conservation begun in the Germany he visited in the 1930s, but, like the DW, it had subsequently gotten buried for decades (Leopold 1936e). The anticipated return of larger birds of prey and predators such as wolf (*Canis lupus*), wildcat (*Felis silvestris*), and lynx (*Lynx lynx*) through migration and release is supported by the ÖJV but still controversial in other circles, where predators are seen as hunting competitors—



Figure 3. Homepage of Aldo Leopold Society for Wildlife Ecology and Nature Conservation, founded in Germany in 1999 (www.ALGnet.de). The badger as a logo is purely incidental, but is intriguing in light of Leopold's affiliation with Wisconsin, the Badger State.

as evidenced by the recent illegal shooting of a lynx in the Bavarian Forest. Leopold would clearly endorse reintroductions, and he would argue that, as in the U.S., remnants of opposition by parts of the hunting and rural public are self-serving and shortsighted, "because you cannot love game and hate predators" (Leopold 1953:146).

Silviculture. Now that the deer problem has been partially solved, gradual silvicultural adjustments are possible in many German forests. Heavy stand-improvement and harvest cuts (mostly shelterwood or selection methods) over large areas are conducted to facilitate natural regeneration and habitat improvement. This increasingly makes superfluous the formerly ubiquitous and expensive deer fences (Figure 4), which in total are estimated to be long enough to reach 3 times around the equator. Where planting for the reintroduction of rare species is still taking place as a transitional tool, such as on the sizable windthrow areas created by the 1990 and 1999 storms, generously spaced hardwoods allow natural establishment of volunteer browse and landscape species of trees and other flora, which until a few years ago were considered undesirable weeds. Such diversification, together with reduced deer pressures, is expected to benefit rare invertebrates and associated birds and mini-wildlife, many of them indicators of environmental health. But game, too, is expected to benefit from less competition in improved habitat conditions. "Fewer but stronger deer," as first promoted by the DW movement in the 1930s (Leopold 1936b:463) and by contemporary quality deer management programs in the U.S., have already been



Figure 4. Monoculturing of conifers and of deer to a great extent account for the lack of diverse ground flora in many German forests. Deer fences have been the standard solution to facilitate natural or artificial regeneration of trees and other flora in the presence of excessive deer populations, as depicted in this scene, where an attempt to reintroduce beech (background) succeeded only with the help of expensive protective measures. Note lack of woody plants outside of fence in the foreground. Photo by H. Schabel.

observed in certain hunting districts, where roe deer weights supposedly have increased by an average of 2 kilograms (Ecological Hunting Club 1999). Rare trees such as species of wild *Malus*, *Pyrus*, *Sorbus*, *Ulmus*, and *Taxus*, among others, are now the focus of programs to map and monitor remaining specimens and of gene banking, propagation, and planting efforts, as aided by public awareness programs. The year 1994, for instance, was dedicated to *Taxus* as “tree of the year.” Leopold the archer and craftsman would have personally identified with that tree, but would have especially applauded this attempt to bring concerns for nature before the general public, as he tried to do himself most of his adult life. Möller (1922:81) had already demanded a “careful search for rare trees and shrubs” and their treatment as “natural monuments,” and in equal spirit Leopold (1936e:102) had considered as “the most pressing job in both Germany and America...to prevent the extermination of rare species.” A recommendation for gradual acceptance of 1 to 10% of trees as snags, in conjunction with resultant large wood debris in forests, also stands to benefit presently still uncommon biota in the more natural forests in the making (Sperber 1994). Initial assessments in forests along gradients of naturalness have already documented beneficial effects of natural

forest management on certain groups of uncommon invertebrates (Bogenschutz 2000).

Concerning timber management, Leopold would heartily endorse the pragmatic positions that seem to emerge from long and at times bitter discussions between nature conservation groups and resource management interests. In his own evolution, Leopold (1991) clearly came to advocate an optimal mix of wild and managed land because, as he put it in a lecture on “Deer and Dauerwald in Germany,” “One cannot divorce esthetics from utility, quality from quantity” (Meine 1988:360). If Leopold (1933:641) argued

that “bread and beauty grow best together” in healthy agricultural landscapes, then he could extend this thinking with Möller’s (1922) argument that wood and beauty need not exclude each other in healthy forests. In this spirit, timber production will, in fact, remain an integral part of forestry in Germany.

Although the global economy allows the importation of wood, Germans believe that high timber inventories and superior timber quality, as recommended by Möller (1922), should remain major objectives for the conversion of existing forests to DW. Once contemporary, storm-related market irregularities have been overcome, lumber prices are anticipated to rise. It is further believed that forfeiting forest production in Germany in favor of pure environmental forestry would only add pressures on already beleaguered forests elsewhere (Moosmayer 1993), a form of NIMBYism similar to the one MacCleery (2000) deplors for contemporary America. Timber production in Germany will, however, be guided by certification procedures. Recently approved standards by the Forest Stewardship Council of Germany (FSC-G) reflect a long-sought compromise reached by an alliance between forestry and most nature protection organizations in that country. These standards fully

accommodate modern DW principles for the future production of "certified" forest products, but they are presently judged too stringent and costly to find broad acceptance among private forest owners (zur Lippe and Baumgarten 2000). Because FSC-G certification requires a clear position with respect to deer-forest priorities (Wald vor Wild), relevant discussions in communal parliaments tend to be dominated more by hunting than any other pertinent issues (Schaefer 2000). It is obviously difficult to reconcile low game densities with maximum hunting income from community-owned hunting leases, and political planning horizons tend to be shallow. As a result, for landowners unwilling to meet FSC standards, alternative certification developed by the German Forestry Council for a Pan European Forest Certification (PEFC) may be more acceptable. PEFC standards, despite being somewhat more lenient with respect to clear-cutting, specifications for pesticide use in forests, and various nature conservation criteria (Sperber, in press), appear to be an acceptable compromise to other than the most dogmatic of nature conservation groups.

Leopold, the wilderness advocate, also would undoubtedly be thrilled to see the various measures in wildlife management and silviculture complemented by serious efforts to create or enhance natural areas, national parks, biosphere reserves, and other protected areas, in addition to the establishment of 5% unmanaged "reference areas" in all state and large communal forests, as approved by the aforementioned FSC-G Working Group (Sperber in press). Any of these efforts are considerably more difficult in a densely populated country without true wilderness than is decreeing protected zones in *de facto* wilderness in a generously spaced setting like North America.

Conclusions

After centuries of commodity-based forest management and almost 100 years of debate, Germany's third attempt to create conditions that promise the sustainability not only of forest products but of forests in the broadest sense seems sufficiently rooted to give hope for durability this time. Contemporary forest policies that mostly evolved since the 1980s put greater weight on ecological processes and on the quality of forests in their totality than on the quantity of preferred ingredients such as deer and wood. They emphasize near-natural

forests that are not so much constructed by man as composed by nature. The important lesson derived from several centuries of experimentation with traditional forest management in the "cradle of scientific forestry" is that the restoration or reinvention of nature may ultimately be the best long-term strategy to regain a healthier environment. Germany arrived at this conclusion the hard way. Following an initially successful reclamation of degraded lands, which averted predicted timber shortages, advancement to the more enlightened stage of the nature- and people-friendly DW took painfully long. Even though it is finally in progress, this transition is as slow as might reasonably be expected in a functional, if cumbersome, pluralistic democracy where compromises between stakeholder groups need to be reached. Points of contention include the fact that a satisfactory reduction of deer numbers has not yet been achieved uniformly throughout the land, pending a more decisive reform of the entrenched hunting system through new federal hunting laws, and that agreement has not been reached on the difficult question of social and ecological carrying capacity. It may well take a generation to incrementally dismantle the last vestiges of historic baggage that perpetuate tensions in the forest-game-nature conservation triad and to implement a game management program less burdened by passions and more guided by science.

While Leopold (1936a:366) would be generally pleased to see some progress concerning his recommendations for a "radical reform in game management...(and)...in silviculture,"—that is, the retreat of conifers and of deer, and a return of biodiversity and beauty—he also would probably see a need to put his observations in the context of Germany's broader environmental health. He would share scientists' great apprehension about the continuing (if easing) air pollution problems, especially with respect to ozone and various forms of nitrogen, as they may involve long-term risks for ecosystem stability that could easily negate most gains in the battle for nature restoration. For Leopold as a Wisconsinite and an American, the general shortage of true nature and the virtual absence of wilderness in densely populated Germany (the wildness of the Autobahn notwithstanding) would run deeply against his grain, even while he would acknowledge that the DW together with newly created protected areas may eventually succeed in recapturing a semblance of nature in parts of that country. Last

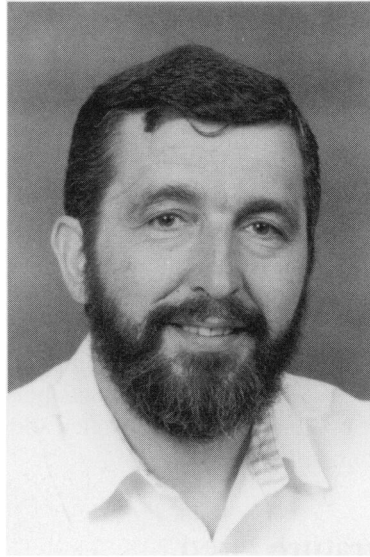
but not least, he would undoubtedly continue to admonish his fellow citizens to cherish, relish, and fight for the generous remnants of nature in America, to be vigilant of the continuing juggernaut of land abuse, and to learn from mistakes of the past, at home as elsewhere.

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